TECHNICAL PROPOSAL · BATTERY ENERGY STORAGE

Solarpark Example Brandenburg

Customer: Example Energy GmbH (fictional) · Site: Brandenburg, Germany (52.35, 14.05) · 01.11.2026
Use case: Recover curtailed PV, shift PV to evening hours, day-ahead arbitrage

Demo data. This report was built with synthetic weather and price data so the tool runs offline. The numbers show how the method works, they are not a real site assessment. Run `bess-designer fetch-data` to use PVGIS and SMARD data.

RECOMMENDATION

30 MW / 63.0 MWh usable BESS

14 × BC-5000 battery containers (70.0 MWh nameplate) and 12 × PCS-2500 power conversion units, AC-coupled at the PV plant's medium-voltage busbar.

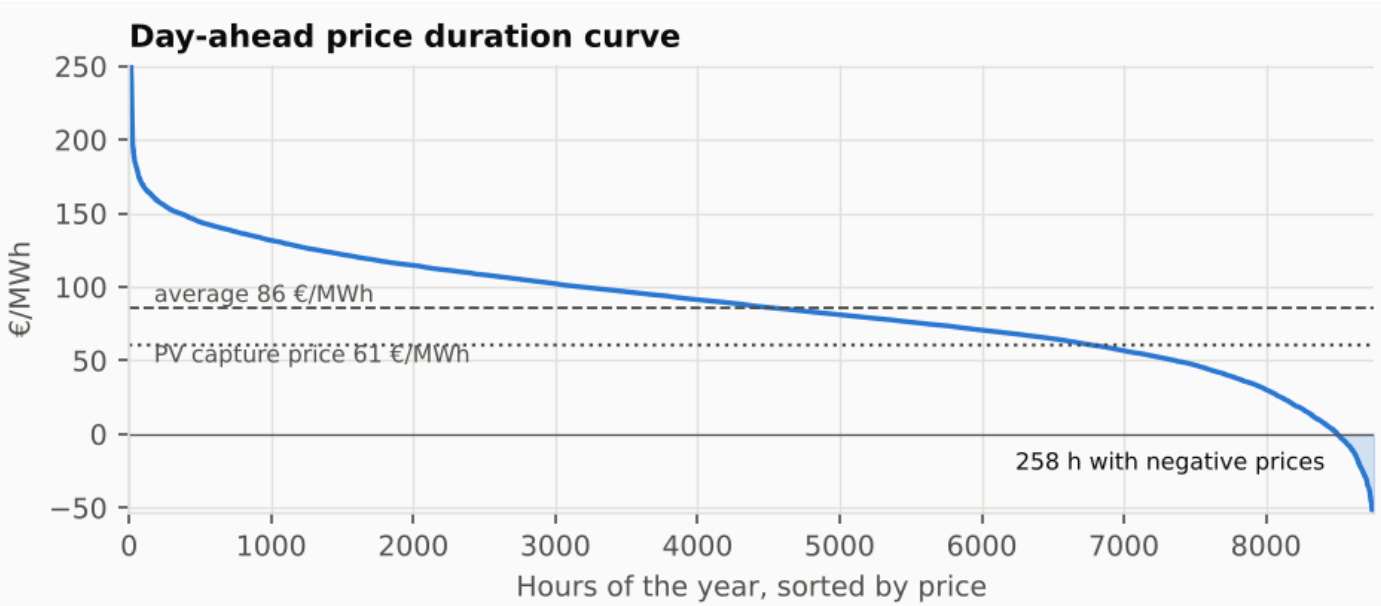
14.7 M€ Investment (CAPEX)	**2 491 k€** Extra revenue, year 1	**3.37 M€** NPV over 15 years	**6.9 a** Simple payback
10.7 % Project IRR	**548** Full cycles per year	**156 MWh** PV curtailment recovered	**126 €** Avg. price when discharging (per MWh)

- 30 MW / 2 h gives the best net present value of the 9 options studied.
- Runner-up is 20 MW / 2 h (NPV 2.34 M€ vs 3.37 M€).
- Revenue per usable MWh and year falls as the duration grows (1 h: 44 k€, 2 h: 40 k€, 4 h: 33 k€), because the extra hours of storage are needed on fewer days. This is why longer systems earn more in total but not more per euro invested.
- Value still grows with power up to the largest size studied. The practical limit is the 30 MW grid connection, which the BESS shares with the PV plant; a larger system would need more grid capacity.
- The recommended system reaches the cycle limit of 1.5 cycles per day, so the warranty terms of the battery directly limit revenue.
- 11 % of the charging energy comes from the PV plant, the rest from the grid in low-price hours. Whether grid charging is allowed for the site (EEG rules for mixed storage) has a large effect; see the sensitivity table.

1. Customer requirements and site

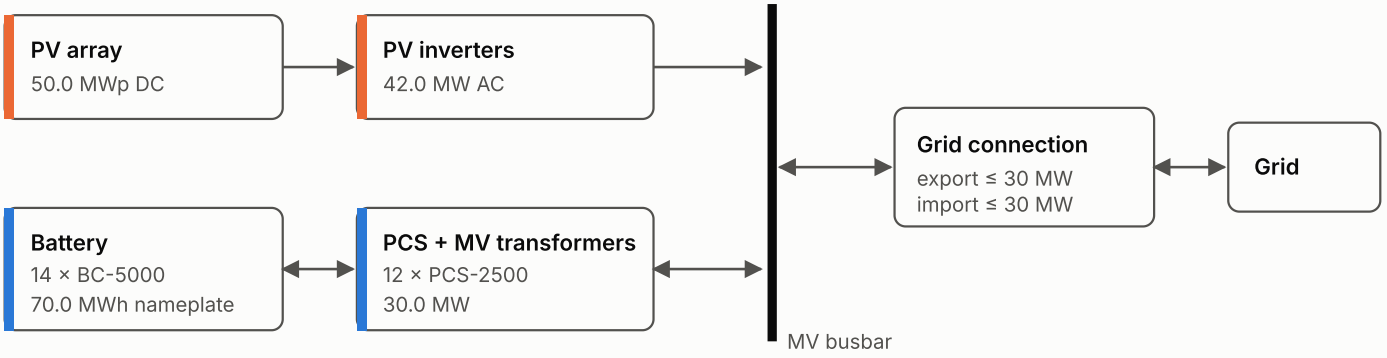
PV plant	50.0 MWp DC / 42.0 MW AC (DC/AC 1.19)
Grid connection (export / import)	30.0 MW / 30.0 MW
Grid charging of the BESS	allowed
PV orientation	tilt 25°, azimuth 180°
Options studied	10, 20, 30 MW × 1, 2, 4 h

PV energy available	47 840 MWh/a (957 kWh/kWp)
Average day-ahead price	86.1 €/MWh
PV capture price (no BESS)	60.9 €/MWh
Hours with negative prices	258 h
PV curtailed without BESS	7 270 MWh/a (7 020 negative prices, 251 grid limit)
PV-only revenue	2.47 M€/a



The PV plant sells mostly in hours when many other PV plants also produce, so it earns less than the average price. The BESS moves energy from those hours to the evening.

2. Proposed solution

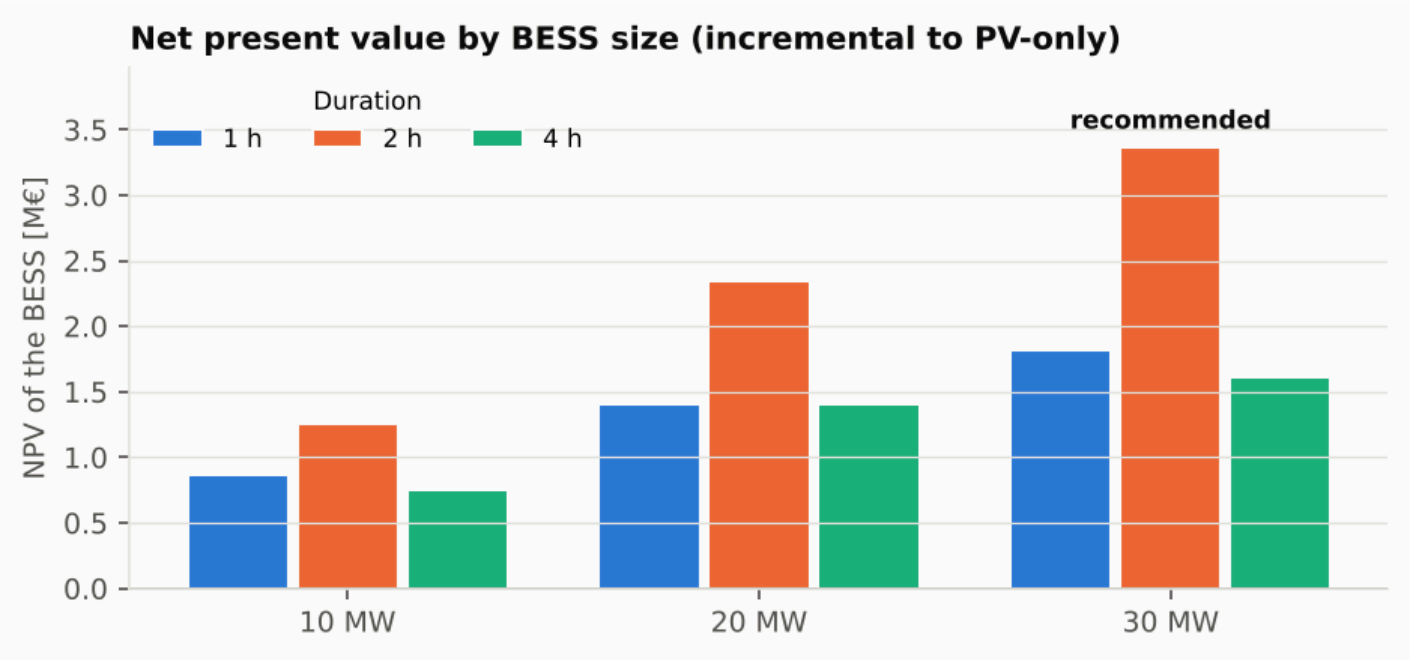


AC-coupled layout: the BESS has its own power conversion and shares the PV plant's grid connection. Export of PV and BESS together is limited to the grid connection capacity.

Main equipment (recommended option)

Item	Type	Qty	Unit rating	Total
Battery container	20 ft LFP battery container, liquid cooled (generic)	14	5.00 MWh	70.0 MWh
Power conversion	Bidirectional PCS skid incl. MV transformer (generic)	12	2.50 MW	30.0 MW

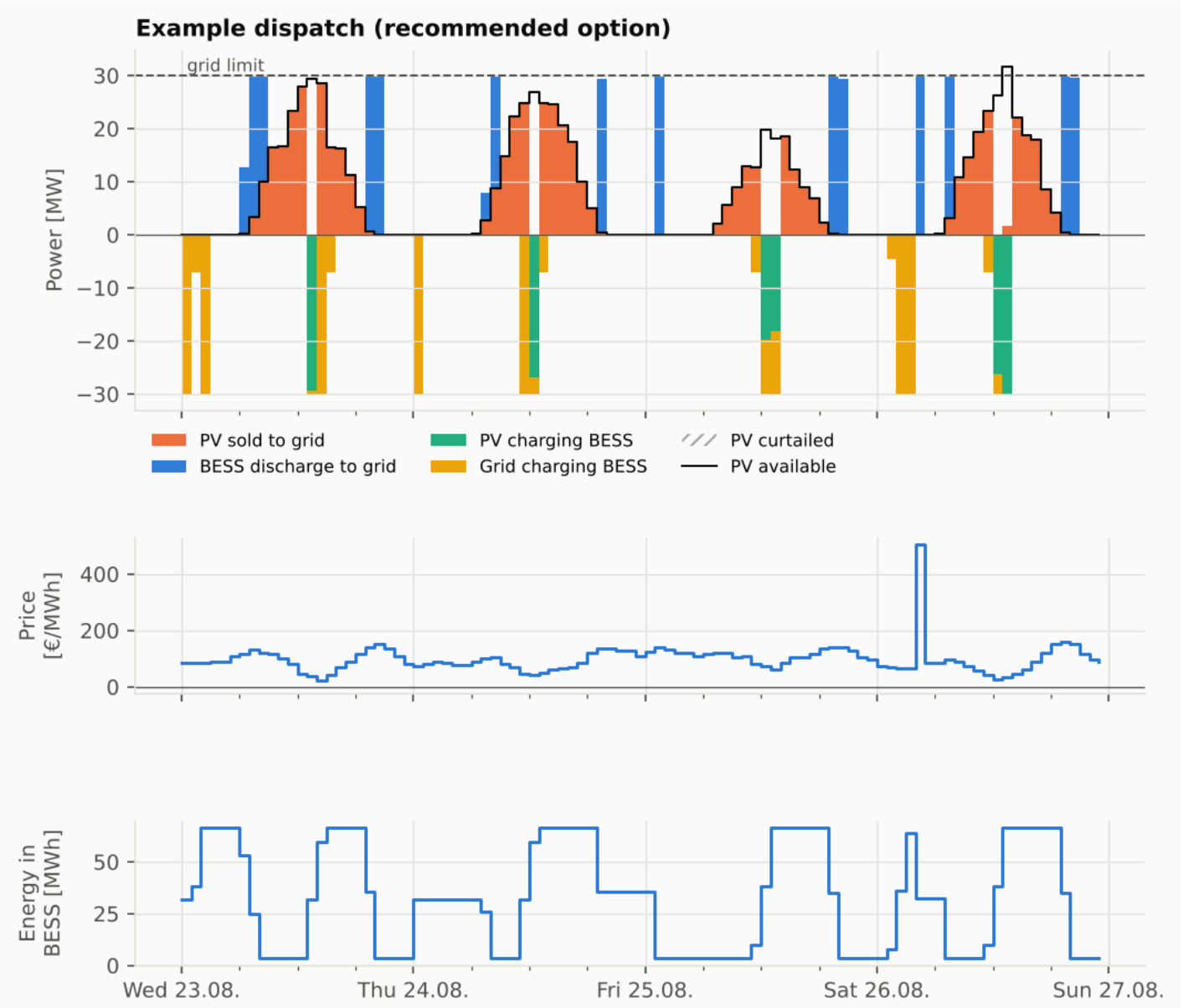
3. Options compared



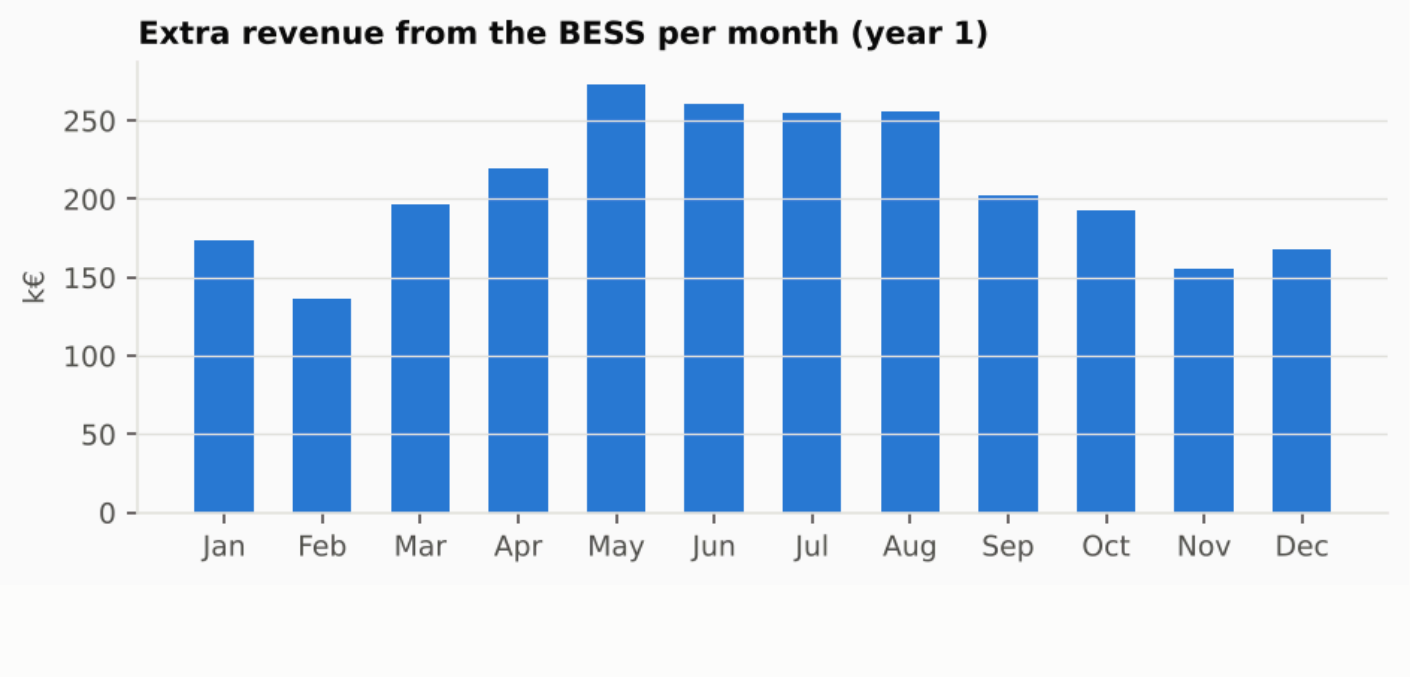
Rank	Option	Usable MWh	CAPEX M€	Extra rev. k€/a	Cycles/a	NPV M€	IRR	Payback a	LCOS €/MWh
1	**30 MW / 2 h**	**63.0**	**14.70**	**2 491**	**548**	**3.37**	**10.7 %**	**6.9**	**60**
2	20 MW / 2 h	40.5	9.50	1 631	548	2.34	11.0 %	6.8	60
3	30 MW / 1 h	31.5	8.40	1 410	548	1.82	10.5 %	7.0	69
4	30 MW / 4 h	121.5	26.40	3 923	503	1.61	8.0 %	8.1	61
5	20 MW / 4 h	81.0	17.60	2 657	514	1.41	8.3 %	8.0	60
6	20 MW / 1 h	22.5	5.90	1 007	547	1.41	10.9 %	6.9	68
7	10 MW / 2 h	22.5	5.20	890	547	1.26	10.9 %	6.8	60
8	10 MW / 1 h	13.3	3.36	581	548	0.86	11.1 %	6.8	65
9	10 MW / 4 h	40.5	8.80	1 335	515	0.76	8.4 %	7.9	60

Usable energy can be larger than power × duration because whole containers are installed. LCOS excludes the cost of the charging energy.

4. How the BESS operates



Above zero: energy fed into the grid. Below zero: energy stored in the battery. Where the PV line is above the orange area, that PV energy goes into the battery or is curtailed. Times in local time (Europe/Berlin).



Energy discharged	34 492 MWh/a
Charged from PV / from grid	4 140 MWh / 35 056 MWh
Average price when charging / discharging	47.3 / 126.0 €/MWh
Model check: simultaneous charge and discharge	0.0 MWh/a (should be close to 0)

5. Sensitivity of the recommendation

Case	Extra rev. k€/a	Cycles/a	NPV M€	Δ NPV M€	IRR	Payback a
Base case	2 491	548	3.37	0.00	10.7 %	6.9
CAPEX -20 %	2 491	548	6.71	3.34	15.8 %	5.3
CAPEX +20 %	2 491	548	0.02	-3.34	7.0 %	8.7
No grid charging (PV only)	1 665	267	-3.29	-6.65	3.0 %	11.6
Max 1 cycle per day	2 236	365	1.31	-2.06	8.5 %	7.9
Round-trip efficiency 0.85	2 390	547	2.55	-0.81	9.9 %	7.3

6. Assumptions and limits of this study

1. Weather and prices of one historical year (2023); data: Synthetic demo data (not real market data).
2. Perfect foresight of day-ahead prices for the whole year. Real trading earns less, so revenues are an upper bound for the day-ahead market alone.
3. Only day-ahead arbitrage and PV shifting are valued. Balancing services (FCR, aFRR), intraday trading and capacity payments are not included.
4. Price taker: the BESS does not change market prices. This makes revenue grow almost linearly with size, which overstates the value of very large systems.
5. AC-coupled BESS. PV energy lost by DC/AC inverter clipping is not recoverable (a DC-coupled design could recover part of it).
6. In hours with negative prices the PV plant is curtailed (no payment for feed-in).
7. Round-trip efficiency 88% (AC to AC), usable state of charge 5% to 95%, max 1.5 cycles per day.
8. Capacity fade 2.0% per year, revenue scales with it. No augmentation.
9. CAPEX 180 €/kWh (nameplate) + 70 €/kW, OPEX 1.5% of CAPEX per year, 15 years, discount rate 7.0%. No taxes, financing or price escalation.
10. Grid charging fee 0 €/MWh.
11. Product catalogue is generic and illustrative; values are not from a specific vendor.

7. Method

For every option the hourly operation of PV plant and BESS is optimised over one year with a linear program (HiGHS solver): maximise market revenue subject to the PV energy balance, the battery energy balance with charge and discharge losses, power and energy limits, the grid connection limit and a daily cycle limit. The same PV plant without storage is the reference case; all values above are the difference to it. Equipment counts come from the product catalogue, and the cash flows over the project lifetime give NPV, IRR, payback and LCOS.